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Battery packaging material having a valve device

Abstract

A battery includes a battery element, housing body, and valve device. The housing body has a laminate including a base material, barrier, and heat-sealable resin layers. The valve device is in communication with the housing body inside. A joined edge portion in which the mutually facing heat-sealable resin layers are fused together is formed in a housing body peripheral edge portion. The valve device includes first and second portions. A valve mechanism reduces the housing body internal pressure if it is increased due to gas generated in the housing body is formed in the first portion. An air passage guides gas generated in the housing body toward the valve mechanism is formed in the second portion. The first portion is located on a joined edge portion edge outer side. At least a portion of the second portion is sandwiched between the heat-sealable resin layers in the joined edge portion.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a divisional application of U.S. application Ser. No. 17/059,903, filed Nov. 30, 2020, the contents of which are incorporated herein by reference.

TECHNICAL FIELD

(1) The present invention relates to a battery.

BACKGROUND ART

(2) JP 2016-152231A (Patent Literature 1) discloses a lithium ion battery that includes a battery element and an enclosing bag that houses the battery element. In this lithium ion battery, a one-way exhaust valve is attached to the enclosing bag. When gas is generated in the enclosing bag as the lithium ion battery is charged or discharged, an excess of the gas is discharged from the one-way exhaust valve (see Patent Literature 1).

CITATION LIST

Patent Literature

(3) Patent Literature 1: JP 2016-152231A

SUMMARY OF INVENTION

Technical Problem

(4) Patent Literature 1 described above does not disclose a method for attaching the one-way exhaust valve to the enclosing bag. If the one-way exhaust valve is attached to the enclosing bag through heat sealing, a valve mechanism within the one-way exhaust valve may break due to heat and pressure that are applied thereto during the heat sealing process.

(5) The present invention was made to solve this problem, and an object of the present invention is to provide a battery in which a valve mechanism is unlikely to break when a valve device is attached to the battery.

Solution to Problem

(6) A battery according to the present invention includes a battery element, a housing body, and a valve device. The housing body is constituted by at least one laminate including at least a base

material layer, a barrier layer, and a heat-sealable resin layer layered in that order and houses the battery element. The valve device is in communication with the inside of the housing body. The heat-sealable resin layers face each other in a peripheral edge portion of the housing body. A joined edge portion in which the mutually facing heat-sealable resin layers are fused together is formed in the peripheral edge portion of the housing body. The valve device includes a first portion and a second portion. A valve mechanism is formed in the first portion, the valve mechanism being configured to reduce an internal pressure of the housing body if the internal pressure is increased due to gas generated in the housing body. An air passage is formed in the second portion, the air passage being configured to guide gas generated in the housing body toward the valve mechanism. The first portion is located on an outer side of an outer edge of the joined edge portion. At least a portion of the second portion is sandwiched between the heat-sealable resin layers in the joined edge portion.

(7) In this battery, the second portion of the valve device is sandwiched between the heat-sealable resin layers in the joined edge portion, but the first portion of the valve device is not sandwiched between the heat-sealable resin layers. Therefore, in this battery, a large pressure and a large amount of heat are not applied to the first portion, when compared to the second portion, when the mutually facing heat-sealable resin layers are fused. As a result, according to this battery, it is possible to keep the valve mechanism in the first portion from breaking due to pressure and heat applied when the mutually facing heat-sealable resin layers are fused.

(8) Preferably, in a thickness direction of the battery, a length of the first portion may be longer than a length of the second portion, and a step may be formed at a boundary between the first portion and the second portion.

(9) In this battery, the first portion is longer than the second portion at least in the thickness direction of the battery, and a step is formed at the boundary between the first portion and the second portion. Accordingly, in this battery, even if the valve device is excessively pressed toward the housing body to sandwich the second portion between the heat-sealable resin layers in a step of manufacturing the battery, the stepped portion is caught on end parts of the laminates. Therefore, according to this battery, it is possible to suppress a situation in which the first portion is sandwiched between the heat-sealable resin layers by mistake in the step of manufacturing the battery. Also, in the thickness direction of the battery described above, the difference between a length of a portion of the joined edge portion where the second portion is sandwiched and a length of a portion of the joined edge portion where the second portion is not sandwiched is small, when compared to a case where the step is not formed at the boundary between the first portion and the second portion. Accordingly, in the portion of the joined edge portion where the second portion is sandwiched, the heat-sealable resin layers are fused together without excessive heat and excessive pressure being applied to the heat-sealable resin layers. As a result, according to this battery, it is possible to suppress a reduction in sealing strength and a reduction in insulating performance, which would be caused if the heat-sealable resin layers become thin. Here, the reduction in the insulating performance refers to a phenomenon in which electricity passes between the barrier (metal) layer and an electrolytic solution as a result of a portion of the heat-sealable resin layers becoming thin or having cracks, for example.

(10) Preferably, a length of the second portion in a width direction of the battery may be longer than a length of the second portion in a thickness direction of the battery.

(11) In this battery, the length of the second portion in the thickness direction of the battery is short, when compared to a case where the cross-sectional shape of the second portion is a perfectly circular shape (the area is the same). That is, in the thickness direction of the battery described above, the difference between a length of a portion of the joined edge portion where the second portion is sandwiched and a length of a portion of the joined edge portion where the second portion is not sandwiched is small. Accordingly, in the portion of the joined edge portion where the second portion is sandwiched, the heat-sealable resin layers are fused together without excessive heat and

excessive pressure being applied to the heat-sealable resin layers. As a result, according to this battery, it is possible to suppress a reduction in the sealing strength and a reduction in the insulating performance, which would be caused if the heat-sealable resin layers become thin.

(12) Preferably, the second portion may include a wing-shaped extended end part that is formed so as to be thinner toward an end part thereof in a width direction of the battery.

(13) In this battery, the length in the thickness direction of the battery smoothly changes from a portion of the joined edge portion where the second portion is not sandwiched to a portion of the joined edge portion where the second portion is sandwiched, when compared to a case where the wing-shaped extended end part is not provided in the second portion. Accordingly, an excessive force is not applied to the laminates at the boundary between a position where the second portion is sandwiched between the heat-sealable resin layers and a position where the second portion is not sandwiched between the heat-sealable resin layers. As a result, according to this battery, the heat-sealable resin layers can be appropriately fused without excessive heat and excessive pressure being applied to the heat-sealable resin layers, and therefore it is possible to suppress a reduction in the sealing strength and a reduction in the insulating performance, which would be caused if the heat-sealable resin layers become thin.

(14) Preferably, a cross-sectional shape of the air passage may be a circular shape.

(15) Preferably, a length of a cross section of the air passage in a width direction of the battery may be longer than a length of the cross section of the air passage in a thickness direction of the battery

(16) Preferably, the second portion may include a pillar that is formed in the air passage.

(17) In this battery, the pillar is formed in the air passage in the second portion, and therefore the air passage is maintained even if pressure and heat are applied to the second portion sandwiched between the mutually facing heat-sealable resin layers. Therefore, according to this battery, the air passage in the second portion can be kept from breaking when the mutually facing heat-sealable resin layers are fused.

(18) Preferably, an outer surface of the second portion may have a satin finish surface.

(19) In this battery, the outer surface of the second portion has a satin finish surface, and accordingly, the heat-sealable resin easily melts at positions where the resin is in contact with the second portion. Therefore, according to this battery, the second portion of the valve device can be more firmly fixed to the housing body, when compared to a case where the outer surface of the second portion is smooth.

(20) Preferably, at least one linear protrusion that extends in a circumferential direction may be formed on an outer surface of the second portion.

(21) The linear protrusion securely comes into contact with the heat-sealable resin layers, and accordingly is easily fused with the laminates. In this battery, the linear protrusion extends in the circumferential direction on the outer surface of the second portion. Therefore, according to this battery, the heat-sealable resin layers and the second portion can be fused in the circumferential direction of the second portion. Also, in this battery, the area of contact between the outer surface of the second portion and the heat-sealable resin is large, when compared to a case where the linear protrusion is not formed on the second portion. Therefore, according to this battery, the second portion of the valve device can be relatively firmly fixed to the housing body. It is also possible to provide a plurality of linear protrusions to more firmly fix the second portion to the housing body.

(22) Preferably, a corner of an end part of the second portion on a side opposite to the first portion is rounded in a plan view.

(23) According to this battery, in a case where the end part on the side opposite to the first portion is located inside the housing body, for example, it is possible to reduce the possibility that the end part will damage the battery element within the housing body. According to this battery, it is also possible to reduce the possibility that the end part will damage the heat-sealable resin layer within the housing body and impair the insulating performance of the heat-sealable resin layer.

(24) Preferably, an external shape of a cross section of the second portion with respect to which a

center line of the air passage is the normal line may be a polygonal shape, and a corner of the polygonal shape may be rounded.

(25) According to this battery, in a case where the end part of the second portion on the side opposite to the first portion is located inside the housing body, it is possible to reduce the possibility that the portion of the second portion located inside the housing body will damage the battery element within the housing body, and reduce the possibility that the portion of the second portion sandwiched between the heat-sealable resin layers will damage the heat-sealable resin layers and impair the insulating performance of the heat-sealable resin layers. Also, according to this battery, in a case where the end part of the second portion on the side opposite to the first portion is sandwiched between the heat-sealable resin layers, it is possible to reduce the possibility that the second portion will damage the heat-sealable resin layers and impair the insulating performance of the heat-sealable resin layers.

(26) Preferably, the first portion and the second portion may be made of different materials, and the material of the first portion may have a higher melting point than the material of the second portion.

(27) In this battery, the material of the first portion has a higher melting point than the material of the second portion, and therefore the first portion is unlikely to deform due to heat even if pressure and heat are applied to the second portion when the mutually facing heat-sealable resin layers are fused. Therefore, according to this battery, the valve mechanism in the first portion can be kept from breaking when the mutually facing heat-sealable resin layers are fused.

(28) Preferably, a flat surface may be formed at least in a portion of an outer surface of at least one of the first portion and the second portion.

(29) In this battery, the flat surface is formed in the outer surface of the valve device, and accordingly, the valve device is prevented from rolling over. According to this battery, the valve device does not roll over when the valve device is to be attached to the housing body, and therefore the valve device can be easily positioned.

(30) Preferably, an adhesive member that is configured to adhere to both the second portion and the heat-sealable resin layers may be arranged between an outer peripheral surface of the second portion and the heat-sealable resin layers.

(31) In this battery, the adhesive member that can adhere to the heat-sealable resin layers is bonded to the outer peripheral surface of the second portion. Therefore, according to this battery, a state where the housing body and the valve device are bonded can be easily maintained irrespective of the material of the second portion.

(32) Preferably, a helium leakage amount from a secondary side to a primary side of the valve device measured in an environment at 25° C. in accordance with a method defined in “vacuum spraying method” of JIS Z2331:2006 “Method for helium leak testing” may be from 5.0×10^{-11} Pa·mL/sec to 5.0×10^{-6} Pa·mL/sec inclusive.

(33) According to this battery, if gas is generated inside the housing body, the gas can be appropriately discharged to the outside of the housing body, and the intrusion of moisture from the external environment into the housing body can be effectively suppressed.

(34) Preferably, a maximum distortion in a thickness direction of the housing body after gas generated in the housing body is discharged via the valve device to the outside of the housing body may be less than 30%.

(35) According to this battery, gas is discharged via the valve device at appropriate timings, and therefore it is possible to reduce the possibility of the occurrence of a situation where large creases are formed in the housing body or the housing body largely deforms after gas is discharged via the valve device to the outside of the housing body.

Advantageous Effects of Invention

(36) According to the present invention, it is possible to provide a battery in which a valve mechanism is unlikely to break when a valve device is attached to the battery.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a plan view of a battery according to a first embodiment;
- (2) FIG. 2 is a cross-sectional view taken along line II-II in FIG. 1;
- (3) FIG. 3 is a diagram showing a housing body;
- (4) FIG. 4 is a diagram showing an example of a cross-sectional structure of packaging materials;
- (5) FIG. 5 is a plan view of a valve device according to the first embodiment;
- (6) FIG. 6 is a cross-sectional view taken along line VI-VI in FIG. 5;
- (7) FIG. 7 is a cross-sectional view taken along line VII-VII in FIG. 5;
- (8) FIG. 8 is a cross-sectional view taken along line VIII-VIII in FIG. 1 and shows an attached state of the valve device;
- (9) FIG. 9 is a flowchart showing a procedure for manufacturing the battery;
- (10) FIG. 10 is a diagram showing an operation of placing the valve device between a flange portion and a packaging material;
- (11) FIG. 11 is a plan view of a valve device according to a second embodiment;
- (12) FIG. 12 is a cross-sectional view taken along line XII-XII in FIG. 11;
- (13) FIG. 13 is a plan view of a valve device according to a third embodiment;
- (14) FIG. 14 is a cross-sectional view taken along line XIV-XIV in FIG. 13;
- (15) FIG. 15 is a plan view of a valve device according to a fourth embodiment;
- (16) FIG. 16 is a cross-sectional view taken along line XVI-XVI in FIG. 15;
- (17) FIG. 17 is a plan view of a valve device according to a fifth embodiment;
- (18) FIG. 18 is a plan view of a valve device according to a sixth embodiment;
- (19) FIG. 19 is a cross-sectional view taken along line XIX-XIX in FIG. 18;
- (20) FIG. 20 is a plan view of a valve device according to a seventh embodiment;
- (21) FIG. 21 is a cross-sectional view taken along line XXI-XXI in FIG. 20;
- (22) FIG. 22 is a diagram showing a state where the valve device is to be attached to the housing body;
- (23) FIG. 23 is a diagram showing a cross section of a valve device according to a first variation;
- (24) FIG. 24 is a diagram showing a cross section of a valve device according to a second variation;
- (25) FIG. 25 is a diagram showing a cross section of a valve device according to a third variation;
- (26) FIG. 26 is a diagram showing a cross section of a valve device according to a fourth variation;
- (27) FIG. 27 is a cross-sectional view taken along line XXVII-XXVII in FIG. 26;
- (28) FIG. 28 is a plan view of a packaging material according to a fifth variation;
- (29) FIG. 29 is a cross-sectional view taken along line XXIX-XXIX in FIG. 28;
- (30) FIG. 30 is a plan view of a valve device according to another variation; and
- (31) FIG. 31 is a cross-sectional view taken along line XXXI-XXXI in FIG. 30.

DESCRIPTION OF EMBODIMENTS

(32) The following describes embodiments of the present invention in detail with reference to the drawings. Note that the same or corresponding portions are denoted with the same reference numerals in the drawings, and descriptions thereof will not be repeated.

1. First Embodiment

(33) 1-1. Overview of Battery

(34) FIG. 1 is a plan view of a battery 10 according to a first embodiment. FIG. 2 is a cross-sectional view taken along line II-II in FIG. 1. In the battery 10, a positive electrode and a negative electrode of tabs 300 are arranged on opposite sides, and the battery 10 is configured to be used, for example, in an electrically driven vehicle such as an electric automobile or a hybrid automobile in which many batteries connected in series are used at a high voltage.

(35) As shown in FIGS. 1 and 2, the battery **10** includes a housing body **100**, a battery element **400**, the tabs **300**, tab films **310**, and a valve device **200**.

(36) The housing body **100** includes packaging materials **110** and **120**. The packaging materials **110** and **120** are subjected to heat sealing in a peripheral edge portion of the housing body **100** to form a joined edge portion **130**. That is, the packaging materials **110** and **120** are fused together in the joined edge portion **130**. Details of the packaging materials **110** and **120** will be described later.

(37) The battery element **400** is a power storage member such as a lithium ion battery or a capacitor. The battery element **400** is housed in the housing body **100**. If an abnormality occurs in the battery element **400**, gas may be generated in the housing body **100**. Also, if the battery element **400** is a capacitor, for example, gas may be generated in the housing body **100** due to a chemical reaction occurring in the capacitor.

(38) The tabs **300** are metal terminals that are used to input and output power to and from the battery element **400**. One end part of each tab **300** is electrically connected to an electrode (a positive electrode or a negative electrode) of the battery element **400**, and the other end part protrudes from an edge of the housing body **100** to the outside.

(39) Examples of metal materials that may constitute the tabs **300** include aluminum, nickel, and copper. If the battery element **400** is a lithium ion battery, for example, a tab **300** that is connected to the positive electrode is usually constituted by aluminum or the like, and a tab **300** that is connected to the negative electrode is usually constituted by copper, nickel, or the like.

(40) The battery **10** includes two tabs **300**. One of the tabs **300** is sandwiched between the packaging materials **110** and **120** via a tab film **310** in an end part of the housing body **100** in the direction of the arrow L. The other tab **300** is sandwiched between the packaging materials **110** and **120** via a tab film **310** in an end part of the housing body **100** in the direction of the arrow R.

(41) The tab films **310** are adhesive protective films and are configured to adhere to both the packaging materials **110** and **120** and the tabs **300** (metal). The tabs **300** made of metal can be fixed with the packaging materials **110** and **120** as a result of the tab films **310** being interposed therebetween. Particularly, in a case where the battery is used at a high voltage, the tab films **310** preferably include a heat-resistant layer or a heat-resistant component and have a function of preventing short circuits.

(42) The valve device **200** is in communication with the inside of the housing body **100** and is configured to discharge gas from the inside to the outside of the housing body **100** if the internal pressure of the housing body **100** has reached or exceeded a predetermined value due to gas generated in the housing body **100**. A housing of the valve device **200** is preferably made of a material that directly adheres to innermost layers of the packaging materials **110** and **120**, and is preferably made of a heat-sealable resin, such as polypropylene (PP), that is the same as the innermost layers of the packaging materials **110** and **120**. In a case where a material other than PP is used for reasons related to heat resistance or the like, it is effective to perform sealing via a film that can adhere to both the other material and PP, similarly to the tab films used for the tabs. An end part of the valve device **200** in the direction of the arrow B is sandwiched between the packaging materials **110** and **120** in an end part of the housing body **100** in the direction of the arrow F. Details of the valve device **200** will be described later.

(43) The battery **10** according to the first embodiment employs various structural features that are devised to attach the valve device **200** to the housing body **100**. The following describes a configuration of the housing body **100**, a configuration of the valve device **200**, a state of the valve device **200** attached to the housing body **100**, and a method for manufacturing the battery **10** in that order.

(44) Note that the directions indicated by arrows L, R, U, D, F, and B are common to the drawings. In the following description, the direction indicated by the arrows L and R will also be referred to as a “width direction of the battery **10**”, and the direction indicated by the arrows U and D will also be referred to as a “thickness direction of the battery **10**”.

(45) 1-2. Configuration of Housing Body

(46) FIG. 3 is a diagram showing the housing body **100**. As shown in FIG. 3, the housing body **100** includes the packaging materials **110** and **120**. The packaging materials **110** and **120** are each constituted by a laminated film and have substantially the same rectangular shape in a plan view.

(47) The packaging material **110** includes a molded part **112** that is molded so as to form a space **S1** and a flange portion **114** that extends in the direction indicated by the arrows **F** and **B** and the direction indicated by the arrows **L** and **R** from the molded part **112**. The molded part **112** is open in a surface in the direction of the arrow **U**. The battery element **400** (FIG. 1) is placed into the space **S1** via the opening in the surface.

(48) FIG. 4 is a diagram showing an example of a cross-sectional structure of the packaging materials **110** and **120**. As shown in FIG. 4, each of the packaging materials **110** and **120** is a laminate constituted by a base material layer **31**, an adhesive agent layer **32**, a barrier layer **33**, an adhesive layer **34**, and a heat-sealable resin layer **35** that are layered in that order. Note that each of the packaging materials **110** and **120** does not necessarily have to include the layers shown in FIG. 4, and is only required to include at least the base material layer **31**, the barrier layer **33**, and the heat-sealable resin layer **35** layered in that order.

(49) The base material layer **31** constitutes the outermost layer of the housing body **100**, and the heat-sealable resin layer **35** constitutes the innermost layer of the housing body **100**. When fabricating the battery **10**, portions of the heat-sealable resin layers **35** located in peripheral edge portions of the packaging materials **110** and **120** are thermally fused together in a state where the battery element **400** (FIG. 2) is arranged in the space **S1** (FIG. 3), whereby the joined edge portion **130** is formed, the battery element **400** is sealed in the housing body **100**, the valve device **200** is fixed to the joined edge portion **130** through fusion, and the tabs **300** are also fixed to the joined edge portion **130** through fusion via the tab films **310**. The following describes the layers included in the packaging materials **110** and **120**. Note that the packaging materials **110** and **120** each have a thickness of about 50 to 200 μm , for example, and preferably about 90 to 150 μm .

(50) 1-2-1. Base Material Layer

(51) The base material layer **31** is a layer that functions as a base material of the packaging material **110** or **120** and constitutes the outermost layer of the housing body **100**.

(52) Materials of the base material layer **31** are not specifically limited so long as the base material layer has an insulating property. Examples of materials of the base material layer **31** include polyesters, polyamides, epoxy resins, acrylic resins, fluorocarbon resins, polyurethanes, silicone resins, phenols, polyetherimides, polyimides, polycarbonates, and mixtures and copolymers thereof. The base material layer **31** may be a resin film made of any of these resins, or may be formed by applying any of these resins. The resin film may be an unstretched film or a stretched film. Examples of stretched films include a uniaxially stretched film and a biaxially stretched film, and the biaxially stretched film is preferable. Examples of stretching methods for forming the biaxially stretched film include a sequential biaxial stretching method, an inflation method, and a simultaneous biaxial stretching method. The base material layer **31** may be a single layer or may be constituted by two or more layers. In a case where the base material layer **31** is constituted by two or more layers, the base material layer **31** may be a laminate that is obtained by stacking resin films using an adhesive or the like, or a laminate that is constituted by two or more co-extruded resin films. A laminate constituted by two or more co-extruded resin films may be used as the base material layer **31** without being stretched, or may be used as the base material layer **31** after being subjected to uniaxial stretching or biaxial stretching. Specific examples of laminates that may be used as the base material layer **31** and are constituted by two or more resin films include a laminate of a polyester film and a nylon film, a laminate of two or more nylon films, and a laminate of two or more polyester films, and the base material layer **31** is preferably a laminate of a stretched nylon film and a stretched polyester film, a laminate of two or more stretched nylon films, or a laminate of two or more stretched polyester films. For example, in a case where the base material layer **31** is

a laminate of two or more resin films, the base material layer **31** is preferably a laminate of a polyester resin film and a polyamide resin film, a laminate of a polyamide resin film and a polyamide resin film, or a laminate of a polyester resin film and a polyamide resin film, and more preferably a laminate of a polyethylene terephthalate film and a polyethylene terephthalate film, a laminate of a nylon film and a nylon film, or a laminate of a polyethylene terephthalate film and a nylon film. Also, it is preferable that a polyester resin constitutes the outermost layer of the base material layer **31**.

(53) The base material layer **31** has a thickness of about 3 to 50 μm , for example, and preferably about 10 to 35 μm .

(54) 1-2-2. Adhesive Agent Layer

(55) The adhesive agent layer **32** is a layer that is arranged on the base material layer **31** as necessary to impart adhesiveness to the base material layer **31**. That is, the adhesive agent layer **32** is provided between the base material layer **31** and the barrier layer **33** as necessary.

(56) The adhesive agent layer **32** is formed of an adhesive that can bond the base material layer **31** and the barrier layer **33** to each other. The adhesive used to form the adhesive agent layer **32** may be a two-liquid curable adhesive or a single-liquid curable adhesive. Also, the bonding mechanism of the adhesive used to form the adhesive agent layer **32** is not specifically limited, and may be any of a chemical reaction mechanism, a solvent volatilizing mechanism, a heat melting mechanism, and a thermocompression bonding mechanism.

(57) The adhesive agent layer **32** has a thickness of about 1 to 10 μm , for example, and preferably about 2 to 5 μm .

(58) 1-2-3. Barrier Layer

(59) The barrier layer **33** is a layer that improves the strength of the packaging material **110** or **120** and prevents water vapor, oxygen, light, and the like from entering the battery **10**. The barrier layer **33** may be constituted by a metal such as aluminum, stainless steel, or titanium, and is preferably constituted by aluminum. The barrier layer **33** can be formed from a metal foil, a vapor-deposited metal film, a vapor-deposited inorganic oxide film, a vapor-deposited carbon-containing inorganic oxide film, or a film provided with any of these vapor-deposited films, for example, and is preferably formed from a metal foil, and more preferably formed from an aluminum foil. In order to prevent the formation of creases or pinholes in the barrier layer **33** when manufacturing the packaging materials, the barrier layer is more preferably formed from a soft aluminum foil made of annealed aluminum (JIS H4160:1994 A8021H-O, JIS H4160:1994 A8079H-O, JIS H4000:2014 A8021P-O, JIS H4000:2014 A8079P-O, etc.), for example.

(60) The thickness of the barrier layer **33** is not specifically limited so long as the barrier layer **33** functions as a barrier layer against water vapor or the like, and may be about 10 to 100 μm , for example, and preferably about 20 to 80 μm .

(61) 1-2-4. Adhesive Layer

(62) The adhesive layer **34** is a layer that is provided between the barrier layer **33** and the heat-sealable resin layer **35** as necessary to firmly bond the heat-sealable resin layer **35**.

(63) The adhesive layer **34** is formed using an adhesive that can bond the barrier layer **33** and the heat-sealable resin layer **35** to each other. Although the composition of the adhesive used to form the adhesive layer **34** is not specifically limited, the composition is, for example, a resin composition that contains an acid-modified polyolefin. Although the acid-modified polyolefin is not specifically limited so long as it is a polyolefin modified with an acid, the acid-modified polyolefin is preferably a polyolefin grafted with an unsaturated carboxylic acid or an anhydride thereof.

(64) The adhesive layer **34** has a thickness of about 1 to 50 μm , for example, and preferably about 2 to 40 μm .

(65) 1-2-5. Heat-Sealable Resin Layer

(66) The heat-sealable resin layer **35** constitutes the innermost layer of the housing body **100**. The

battery element **400** is sealed in the housing body **100** by thermally fusing mutually facing heat-sealable resin layers **35** in the peripheral edge portion of the housing body **100**. Also, insulation quality can be maintained between an electrolytic solution and a metal constituting the barrier layer as a result of the barrier layer being covered with the heat-sealable resin layer that has at least a predetermined thickness.

(67) Although a resin component that is used for the heat-sealable resin layer **35** is not specifically limited so long as the heat-sealable resin layer **35** can be thermally fused, the resin component is a polyolefin or an acid-modified polyolefin, for example.

(68) Examples of polyolefins include: polyethylenes such as low-density polyethylene, medium-density polyethylene, high-density polyethylene, and linear low-density polyethylene; crystalline or noncrystalline polypropylenes such as homopolypropylene, block copolymers of polypropylene (e.g., block copolymer of propylene and ethylene), and random copolymers of polypropylene (e.g., random copolymer of propylene and ethylene); and a terpolymer of ethylene, butene, and propylene. Among these polyolefins, polyethylenes and polypropylenes are preferable. Also, although the acid-modified polyolefin is not specifically limited so long as it is a polyolefin modified with an acid, the acid-modified polyolefin is preferably a polyolefin grafted with an unsaturated carboxylic acid or an anhydride thereof.

(69) The thickness of the heat-sealable resin layer **35** is not specifically limited, but preferably 100 μm or less, more preferably about 15 to 90 μm , and further preferably about 30 to 80 μm .

(70) 1-3. Configuration of Valve Device

(71) FIG. 5 is a plan view of the valve device **200**. As shown in FIG. 5, the valve device **200** includes a valve function portion **210** and a seal attachment portion **220**. Although details will be described later, at least a portion of the seal attachment portion **220** is fixed in a state of being sandwiched between the packaging materials **110** and **120** (FIG. 2), and an outer peripheral surface of the seal attachment portion **220** and the heat-sealable resin layers **35**, which are the innermost layers of the packaging materials **110** and **120**, are joined as a result of being fused through heat sealing.

(72) Rs are formed at corners of an end part of the seal attachment portion **220** in the direction of the arrow B. That is, Rs (e.g., $R=0.2\text{ mm}$ to 2.0 mm) are formed at corners of an end part of the seal attachment portion **220** on the side opposite to the valve function portion **210** in a plan view. Note that in the present specification, the expression “an R being formed” means that a corner is rounded. Here, in the description of a structure, the expression “an R being formed” means that a corner is rounded similarly to the case where the corner is chamfered, and where “R” is described alone, “R” represents the radius of curvature of the corner. Note that it is possible to chamfer sharp corners that are formed in a step of manufacturing the valve device **200** to make the corners round (form Rs), but if the housing of the valve device **200** is a resin molded article, Rs can also be formed by molding the housing so as to originally include rounded corners, without performing chamfering such as cutting.

(73) FIG. 6 is a cross-sectional view taken along line VI-VI in FIG. 5. As shown in FIG. 6, cross sections of the valve function portion **210** and the seal attachment portion **220** of the valve device **200** have perfectly circular shapes, and an air passage **A1** is formed inside the seal attachment portion **220**. A cross section of the air passage **A1** has a perfectly circular shape.

(74) In the valve device **200**, a length **L2** of the valve function portion **210** in the thickness direction (the direction indicated by the arrows U and D) of the battery **10** is longer than a length **L1** of the seal attachment portion **220** in the thickness direction of the battery **10**. A length **L2** of the valve function portion **210** in the width direction (the direction indicated by the arrows L and R) of the battery **10** is longer than a length **L1** of the seal attachment portion **220** in the width direction of the battery **10**. That is, the diameter of the cross section of the valve function portion **210** is longer than the diameter of the cross section of the seal attachment portion **220**. As a result, a step is formed at the boundary between the valve function portion **210** and the seal attachment portion **220**

(FIG. 5).

(75) FIG. 7 is a cross-sectional view taken along line VII-VII in FIG. 5. As shown in FIG. 7, Rs (e.g., $R=0.2\text{ mm to }2.0\text{ mm}$) are formed in the end part of the seal attachment portion **220** in the direction of the arrow B. Also, the air passage **A1** is formed inside the seal attachment portion **220**. The air passage **A1** guides gas generated in the housing body **100** toward the valve function portion **210**, for example.

(76) A valve mechanism that is configured to discharge gas generated in the housing body **100** (FIG. 1) is provided inside the valve function portion **210**. Specifically, the valve function portion **210** includes an O ring **212**, a ball **214**, a spring **216**, and a membrane **218**. That is, a ball spring valve mechanism is provided in the valve function portion **210**. Note that the valve mechanism provided in the valve function portion **210** is not specifically limited so long as the mechanism can reduce the internal pressure of the housing body **100** that has been increased by gas, and may also be a poppet valve mechanism, a duckbill valve mechanism, an umbrella valve mechanism, or a diaphragm valve mechanism, for example.

(77) The O ring **212** is a hollow circular ring and is made of fluororubber, for example. The ball **214** and the spring **216** are made of stainless steel, for example. Note that the ball **214** may also be made of resin. The membrane **218** has a pore diameter of about $10.\text{sup.}-2\text{ to }10.\text{sup.}0\text{ }\mu\text{m}$, for example, and is constituted by a PTFE membrane that keeps the electrolytic solution from leaking and allows only gas to permeate (selective permeation). Note that PTFE means polytetrafluoroethylene. Since the PTFE membrane is a soft material, if the strength of the membrane is insufficient, it is also possible to form the membrane and a mesh or a non-woven cloth of polypropylene, polyester, or the like into a single piece and use the thus reinforced membrane.

(78) If the internal pressure of the housing body **100** reaches a predetermined pressure in a state where the valve device **200** is attached to the housing body **100**, gas guided through the air passage **A1** presses the ball **214** in the direction of the arrow F. As a result of the ball **214** being pressed and the spring **216** being compressed, the gas inside the housing body **100** passes through a gap that is formed between the ball **214** and the O ring **212**, permeates the membrane **218**, and is discharged from an exhaust port **O1** to the outside of the housing body **100**.

(79) 1-4. Attached State of Valve Device

(80) FIG. 8 is a cross-sectional view taken along line VIII-VIII in FIG. 1 and shows an attached state of the valve device **200**. As shown in FIG. 8, the valve function portion **210** of the valve device **200** is located on an outer side of an edge of the joined edge portion **130**. On the other hand, a portion of the seal attachment portion **220** of the valve device **200** is sandwiched between the heat-sealable resin layer **35** of the packaging material **110** and the heat-sealable resin layer **35** of the packaging material **120** in the joined edge portion **130**, and the outer peripheral surface of the seal attachment portion **220** and the heat-sealable resin layers **35** constituting the innermost layers of the packaging materials **110** and **120** are joined through fusion. Note that for the sake of convenience, FIG. 8 partially shows the heat-sealable resin layers **35** only in the vicinity of the joined edge portion **130** to show the state where the valve device **200** is joined through fusion with the heat-sealable resin layers **35**, which are the innermost layers of the packaging materials **110** and **120**, but the heat-sealable resin layers **35** are provided over all the surfaces of the packaging materials **110** and **120**.

(81) The following describes reasons why the seal attachment portion **220** is sandwiched between the heat-sealable resin layers **35** in the joined edge portion **130** and the valve function portion **210** is not sandwiched between the heat-sealable resin layers **35** in the joined edge portion **130** in the battery **10** according to the first embodiment.

(82) Assume that the valve function portion **210** is sandwiched between the heat-sealable resin layers **35** in the joined edge portion **130**. In this case, the valve mechanism in the valve function portion **210** may break due to heat and pressure that are applied when the heat-sealable resin layers

35 are fused together (subjected to heat sealing) in the peripheral edge portions of the packaging materials **110** and **120**.

(83) In the battery **10** according to the first embodiment, the seal attachment portion **220** is sandwiched between the heat-sealable resin layers **35** in the joined edge portion **130**, but the valve function portion **210** is not sandwiched between the heat-sealable resin layers **35**. Therefore, in the battery **10**, a large pressure and a large amount of heat are not applied to the valve function portion **210** during the heat sealing process. That is, in the battery **10**, the valve function portion **210** is not sandwiched between the heat-sealable resin layers **35** to keep the valve mechanism from breaking due to pressure and heat applied during the heat sealing process.

(84) Also, in the battery **10** according to the first embodiment, the cross section of the seal attachment portion **220** has a smaller diameter than the cross section of the valve function portion **210** as described above. Accordingly, in the thickness direction of the battery, the difference between a length **L4** of a portion of the joined edge portion **130** where the seal attachment portion **220** is sandwiched and a length **L3** of a portion of the joined edge portion **130** where the seal attachment portion **220** is not sandwiched is small, when compared to a case where the diameter of the cross section of the seal attachment portion **220** is larger than or equal to the diameter of the cross section of the valve function portion **210**. The larger the difference is, the more pressure applied during the heat sealing process needs to be increased to join the outer peripheral surface of the seal attachment portion **220** and the heat-sealable resin layers constituting the innermost layers of the packaging materials **110** and **120** without forming a gap therebetween. As a result, pressure applied to the peripheral edge portion of the housing body **100** during the heat sealing process is increased. If the pressure is increased, the heat-sealable resin layers **35** may become thin particularly at the position where the seal attachment portion **220** is sandwiched, and also at positions where the tab films **310** and the tabs **300** are sandwiched. If the heat-sealable resin layers **35** become thin, a dielectric breakdown may occur in the battery **10**.

(85) In the battery **10** according to the first embodiment, the difference between the length **L4** and the length **L3** is small as described above. Accordingly, when the peripheral edge portion of the housing body **100** is sandwiched by a heat sealing apparatus, pressure and heat are appropriately applied to the heat-sealable resin layers over the entire periphery of the housing body **100**. As a result, according to the battery **10**, it is possible to appropriately fuse the mutually facing heat-sealable resin layers **35** and firmly fix the seal attachment portion **220** to the housing body **100** while reducing the possibility of a dielectric breakdown in the battery **10**.

(86) Also, in the battery **10** according to the first embodiment, the end part of the seal attachment portion **220** in the direction of the arrow **B** protrudes into the space **S1** past the flange portion **114**. Therefore, depending on conditions in which the battery **10** is used, the end part of the seal attachment portion **220** in the direction of the arrow **B** may come into contact with the battery element **400**. In the battery **10** according to the first embodiment, **Rs** are formed in the end part of the seal attachment portion **220** in the direction of the arrow **B** as described above (FIG. 5). Therefore, even if the end part of the seal attachment portion **220** comes into contact with the battery element **400**, the end part is unlikely to damage the battery element **400**. Also, depending on conditions in which the battery **10** is used, the end part of the seal attachment portion **220** in the direction of the arrow **B** may come into contact with the heat-sealable resin layer **35** of the packaging material **120**. In the battery **10** according to the first embodiment, **Rs** are formed in the end part of the seal attachment portion **220** in the direction of the arrow **B** as described above, and therefore, even if the end part of the seal attachment portion **220** comes into contact with the heat-sealable resin layer **35** of the packaging material **120**, the end part is unlikely to damage the heat-sealable resin layer **35**.

(87) 1-5. Manufacturing Method

(88) FIG. 9 is a flowchart showing a procedure for manufacturing the battery **10**. The battery **10** is manufactured using a manufacturing apparatus, for example.

(89) As shown in FIG. 9, the manufacturing apparatus places components in the housing body **100** (step **S100**). For example, the manufacturing apparatus places the battery element **400**, to which the tabs **300** provided with the tab films **310** have been electrically connected through welding, into the space **S1** within the packaging material **110** such that the tabs provided with the tab films **310** are placed on the flange portion **114** of the packaging material **110**, and then places the valve device **200** on the flange portion **114** of the packaging material **110**. Note that it is also possible to place the battery element **400** into the space **S1** within the packaging material **110**, then electrically connect the tabs **300** provided with the tab films **310** to the battery element **400** through welding such that the tabs and the tab films **310** are placed on the flange portion **114** of the packaging material **110**, and then place the valve device **200** on the flange portion **114** of the packaging material **110**. Then, the manufacturing apparatus places the packaging material **120** on the packaging material **110**.

(90) FIG. 10 is a diagram showing an operation of placing the valve device **200** between the flange portion **114** of the packaging material **110** and the packaging material **120**. As shown in FIG. 10, a step is formed between the valve function portion **210** and the seal attachment portion **220**.

Accordingly, even if the valve device **200** is excessively pressed toward the housing body **100** to sandwich the seal attachment portion **220** between the packaging materials **110** and **120**, the stepped portion is caught on end parts of the packaging materials **110** and **120**. Therefore, according to the battery **10**, it is possible to suppress a situation in which the valve function portion **210** is sandwiched between the packaging materials **110** and **120** (the heat-sealable resin layers **35**) by mistake in the step of manufacturing the battery **10**.

(91) After the components are placed, the manufacturing apparatus performs heat sealing on the peripheral edge portion of the housing body **100** (step **S110**). That is, the manufacturing apparatus sandwiches the peripheral edge portion of the housing body **100** and applies pressure and heat to the peripheral edge portion of the housing body **100**. As a result, the mutually facing heat-sealable resin layers **35** are fused together in the peripheral edge portion of the housing body **100**, and the joined edge portion **130** is formed. Also, the battery element **400** is sealed in the housing body **100**, the valve device **200** is fixed to the joined edge portion **130** through fusion, and the tabs **300** are also fixed to the joined edge portion **130** through fusion via the tab films **310**, whereby the battery **10** is complete. Note that in the heat sealing step, gas is removed from the inside of the housing body **100** to achieve a state where unneeded gas is not contained inside the housing body **100**.

Specifically, a portion of the periphery of the housing body **100** is left unjoined, without the entire periphery being joined, gas is removed via the unjoined portion of the periphery, and finally pressure and heat are applied to the unjoined portion of the periphery to complete the joined edge portion **130** over the entire periphery. In the case of a battery that requires an electrolytic solution, a portion of the periphery of the housing body **100** is left unjoined, without the entire periphery being joined, the electrolytic solution is introduced via the unjoined portion of the periphery, gas is removed, and finally pressure and heat are applied to the unjoined portion of the periphery to complete the joined edge portion **130** over the entire periphery.

(92) Also, it is effective to make surfaces of seal bars of the manufacturing apparatus that sandwich the peripheral edge portion of the housing body **100** have a shape that conforms to the external shape of the seal attachment portion **220**. In this case, the heat-sealable resin layers **35** are more firmly joined together at the position where the seal attachment portion **220** is sandwiched. In this case as well, it is effective to make the seal attachment portion **220** have a flattened shape as in a second embodiment described below, in order to suppress deformation of the packaging materials **110** and **120** and a load applied thereto.

(93) 1-6. Features

(94) As described above, in the battery **10** according to the first embodiment, at least a portion of the seal attachment portion **220** of the valve device **200** is sandwiched between the heat-sealable resin layers **35** in the joined edge portion **130**, and the valve function portion **210** of the valve

device **200** is not sandwiched between the heat-sealable resin layers **35** in the joined edge portion **130**. Therefore, in the battery **10**, a large pressure and a large amount of heat are not applied to the valve function portion **210**, when compared to the seal attachment portion **220**, when the mutually facing heat-sealable resin layers **35** are fused. As a result, according to the battery **10**, it is possible to keep the valve mechanism in the valve function portion **210** from breaking due to pressure and heat applied when the mutually facing heat-sealable resin layers **35** are fused.

(95) Note that the battery element **400** is an example of a “battery element” in the present invention, the housing body **100** is an example of a “housing body” in the present invention, and the valve device **200** is an example of a “valve device” in the present invention. The base material layer **31** is an example of a “base material layer” in the present invention, the barrier layer **33** is an example of a “barrier layer” in the present invention, and the heat-sealable resin layer **35** is an example of a “heat-sealable resin layer” in the present invention. The joined edge portion **130** is an example of a “joined edge portion” in the present invention. The valve function portion **210** is an example of a “first portion” in the present invention, and the seal attachment portion **220** is an example of a “second portion” in the present invention. The air passage **A1** is an example of an “air passage” in the present invention.

(96) For the sake of convenience, the size of the battery element **400** is shown as being small relative to the space **S1** within the housing body **100** in order to make it easy to understand that the battery element **400** is housed in the space **S1** within the housing body **100**. Although the space **S1** is slightly larger than the battery element **400** in order to place the battery element **400** into the space **S1** in a manufacturing step, gas is removed in a manufacturing step as described above, and accordingly, the space **S1** is slightly reduced as the gas is removed, and in a final state of the battery **10**, the space **S1** has substantially the same size as the battery element **400**, and the battery element **400** is housed in the space **S1** with almost no gap formed therebetween.

2. Second Embodiment

(97) The second embodiment differs from the first embodiment described above in the configuration of the valve device. Other configurations are basically the same as those in the first embodiment. Here, differences from the first embodiment will be described.

(98) FIG. **11** is a plan view of a valve device **200A** that is mounted in a battery according to the second embodiment. As shown in FIG. **11**, the valve device **200A** includes a valve function portion **210A** and a seal attachment portion **220A**. At least a portion of the seal attachment portion **220A** is sandwiched between the packaging materials **110** and **120** and is subjected to heat sealing. The cross-sectional shape of the seal attachment portion **220A** differs from that in the first embodiment. The valve function portion **210A** is basically the same as that in the first embodiment, but the shapes of the housing and the valve mechanism are partially changed since an air passage **A6** (FIG. **12**) formed in the seal attachment portion **220A** has a different shape.

(99) FIG. **12** is a cross-sectional view taken along line XII-XII in FIG. **11**. As shown in FIG. **12**, in a cross section of the seal attachment portion **220A**, a length **L5** in the width direction (the direction indicated by the arrows **L** and **R**) of the battery is longer than a length **L6** in the thickness direction (the direction indicated by the arrows **U** and **D**) of the battery. More specifically, the cross-sectional shape of the seal attachment portion **220A** is an elliptical shape.

(100) The air passage **A6** is formed inside the seal attachment portion **220A**. The length of the air passage **A6** in the width direction of the battery is longer than the length of the air passage **A6** in the thickness direction of the battery. More specifically, the cross-sectional shape of the air passage **A6** is an elliptical shape.

(101) As described above, in the second embodiment, the length **L5** of the cross section of the seal attachment portion **220A** in the width direction of the battery is longer than the length **L6** of the cross section of the seal attachment portion **220A** in the thickness direction of the battery. That is, the length of the seal attachment portion **220A** in the thickness direction of the battery is short, when compared to a case where the cross-sectional shape of the seal attachment portion is a

perfectly circular shape (the area is the same). In the thickness direction of the battery described above, the difference between a length of a portion of the joined edge portion **130** where the seal attachment portion **220A** is sandwiched and a length of a portion of the joined edge portion **130** where the seal attachment portion **220A** is not sandwiched is further reduced. Therefore, according to this battery, it is possible to appropriately apply pressure and heat to the heat-sealable resin layers **35** over the entire periphery of the housing body **100** and appropriately fuse the mutually facing heat-sealable resin layers **35**, and therefore the seal attachment portion **220A** of the valve device **200A** can be firmly fixed to the housing body **100**.

(102) Note that the valve device **200A** is an example of the “valve device” in the present invention, the valve function portion **210A** is an example of the “first portion” in the present invention, and the seal attachment portion **220A** is an example of the “second portion” in the present invention. The air passage **A6** is an example of the “air passage” in the present invention.

3. Third Embodiment

(103) A third embodiment differs from the first embodiment described above in the configuration of the valve device. Other configurations are basically the same as those in the first embodiment. Here, differences from the first embodiment will be described.

(104) FIG. **13** is a plan view of a valve device **200B** that is mounted in a battery according to the third embodiment. As shown in FIG. **13**, the valve device **200B** includes a valve function portion **210B** and a seal attachment portion **220B**. At least a portion of the seal attachment portion **220B** is sandwiched between the packaging materials **110** and **120** and is subjected to heat sealing. The cross-sectional shape of the seal attachment portion **220B** differs from that in the first embodiment. The valve function portion **210B** is basically the same as that in the first embodiment, but the shapes of the housing and the valve mechanism are partially changed since an air passage **A7** (FIG. **14**) formed in the seal attachment portion **220B** has a different shape.

(105) FIG. **14** is a cross-sectional view taken along line XIV-XIV in FIG. **13**. As shown in FIG. **14**, wing-shaped extended end parts **40** and **41** are formed on both end parts of the seal attachment portion **220B** in the width direction (the direction indicated by the arrows L and R) of the battery. The wing-shaped extended end parts **40** and **41** are formed so as to be thinner toward the ends thereof in the width direction of the battery. In other words, it could be said that the wing-shaped extended end parts **40** and **41** are portions in which the length thereof in the thickness direction of the battery gently changes in the direction indicated by the arrows L and R, when compared to the other portion (circular portion) of the seal attachment portion **22**.

(106) In the battery according to the third embodiment, the length in the thickness direction of the battery smoothly changes from a portion of the joined edge portion **130** where the seal attachment portion **220B** is not sandwiched to a portion of the joined edge portion **130** where the seal attachment portion **220B** is sandwiched, when compared to the first embodiment (a case where the wing-shaped extended end parts **40** and **41** are not provided in the seal attachment portion **220B**). Accordingly, in this battery, an excessive force is not applied to the packaging materials **110** and **120** at the boundary between a position where the seal attachment portion **220B** is sandwiched between the heat-sealable resin layers **35** and a position where the seal attachment portion **220B** is not sandwiched between the heat-sealable resin layers **35**, and therefore the seal attachment portion **220B** of the valve device **200B** can be firmly fixed to the housing body **100**.

(107) Note that the valve device **200B** is an example of the “valve device” in the present invention, the valve function portion **210B** is an example of the “first portion” in the present invention, and the seal attachment portion **220B** is an example of the “second portion” in the present invention. The wing-shaped extended end parts **40** and **41** are each an example of a “wing-shaped extended end part” in the present invention. The air passage **A7** is an example of the “air passage” in the present invention.

4. Fourth Embodiment

(108) A fourth embodiment differs from the first embodiment described above in the configuration

of the valve device. Other configurations are basically the same as those in the first embodiment. Here, differences from the first embodiment will be described.

(109) FIG. 15 is a plan view of a valve device **200C** that is mounted in a battery according to the fourth embodiment. As shown in FIG. 15, the valve device **200C** includes a valve function portion **210C** and a seal attachment portion **220C**. At least a portion of the seal attachment portion **220C** is sandwiched between the packaging materials **110** and **120** and is subjected to heat sealing. The cross-sectional shape of the seal attachment portion **220C** differs from that in the first embodiment. The valve function portion **210C** is basically the same as that in the first embodiment, but the shapes of the housing and the valve mechanism are partially changed since an air passage **A2** (FIG. 16) formed in the seal attachment portion **220C** has a different shape.

(110) FIG. 16 is a cross-sectional view taken along line XVI-XVI in FIG. 15. As shown in FIG. 16, pillars **50** and **51** are formed inside the seal attachment portion **220C** (in the air passage **A2**). The pillars **50** and **51** each extend in the thickness direction (the direction indicated by the arrows **U** and **D**) of the battery, and both ends of each pillar in the thickness direction of the battery are connected to an inner peripheral surface of the seal attachment portion **220C**. Also, the pillars **50** and **51** each extend in the direction indicated by the arrows **F** and **B** in the air passage **A2** (FIG. 15). Note that the number of pillars does not necessarily have to be two, and it is sufficient that at least one pillar is provided.

(111) In the battery according to the fourth embodiment, the pillars **50** and **51** are formed in the air passage **A2**, and therefore, the air passage **A2** is maintained even if pressure and heat are applied to the seal attachment portion **220C** sandwiched between the mutually facing heat-sealable resin layers **35**. Therefore, according to this battery, the air passage **A2** in the seal attachment portion **220C** can be kept from breaking when the mutually facing heat-sealable resin layers **35** are fused.

(112) Note that the valve device **200C** is an example of the “valve device” in the present invention, the valve function portion **210C** is an example of the “first portion” in the present invention, and the seal attachment portion **220C** is an example of the “second portion” in the present invention. The pillars **50** and **51** are each an example of a “pillar” in the present invention. The air passage **A2** is an example of the “air passage” in the present invention.

5. Fifth Embodiment

(113) A fifth embodiment differs from the first embodiment described above in the configuration of the valve device. Other configurations are basically the same as those in the first embodiment. Here, differences from the first embodiment will be described.

(114) FIG. 17 is a plan view of a valve device **200D** that is mounted in a battery according to the fifth embodiment. As shown in FIG. 17, the valve device **200D** includes a valve function portion **210** and a seal attachment portion **220D**. The configuration of the valve function portion **210** is the same as that in the first embodiment.

(115) At least a portion of the seal attachment portion **220D** is sandwiched between the packaging materials **110** and **120** and is subjected to heat sealing. The outer surface of the seal attachment portion **220D** differs from that in the first embodiment. Specifically, the outer surface of the seal attachment portion **220D** has a satin finish surface. The satin finish surface has a surface roughness **Ra** of 1 μm to 20 μm , for example.

(116) In the battery according to the fifth embodiment, the outer surface of the seal attachment portion **220D** has a satin finish surface, and accordingly, the heat-sealable resin easily melts at positions where the resin is in contact with the seal attachment portion **220D**. Therefore, according to this battery, the seal attachment portion **220D** of the valve device **200D** can be more firmly fixed to the housing body **100**, when compared to the first embodiment (a case where the outer surface of the seal attachment portion **220D** is smooth).

(117) Note that the valve device **200D** is an example of the “valve device” in the present invention, and the seal attachment portion **220D** is an example of the “second portion” in the present invention.

6. Sixth Embodiment

(118) A sixth embodiment differs from the first embodiment described above in the configuration of the valve device. Other configurations are basically the same as those in the first embodiment. Here, differences from the first embodiment will be described.

(119) FIG. **18** is a plan view of a valve device **200E** that is mounted in a battery according to the sixth embodiment. As shown in FIG. **18**, the valve device **200E** includes a valve function portion **210** and a seal attachment portion **220E**. The configuration of the valve function portion **210** is the same as that in the first embodiment.

(120) At least a portion of the seal attachment portion **220E** is sandwiched between the packaging materials **110** and **120** and is subjected to heat sealing. The outer surface of the seal attachment portion **220E** differs from that in the first embodiment. Specifically, linear protrusions **60** that continuously extend over the whole circumference of the seal attachment portion **220E** are formed on the outer surface. Three linear protrusions **60** are formed in the direction indicated by the arrows F and B in the seal attachment portion **220E**. Note that the number of linear protrusions **60** does not necessarily have to be three, and it is sufficient that at least one linear protrusion is provided.

(121) FIG. **19** is a cross-sectional view taken along line XIX-XIX in FIG. **18**. As shown in FIG. **19**, a cross section of each linear protrusion **60** has a semicircular shape. R of the semicircular shape is 0.05 mm to 1.0 mm, for example. A diameter L12 (the length in the thickness direction of the battery and the length in the width direction of the battery) of a portion of the seal attachment portion **220E** where the linear protrusion **60** is formed is larger than a diameter L11 of a portion of the seal attachment portion **220E** where the linear protrusion **60** is not formed.

(122) The linear protrusions **60** securely come into contact with the heat-sealable resin layers **35** during the heat sealing process, and accordingly, are easily fused with the packaging materials **110** and **120**. In the battery according to the sixth embodiment, the linear protrusions **60** continuously extend over the whole circumference of the seal attachment portion **220E** on its outer surface. Therefore, according to this battery, the heat-sealable resin layers **35** and the seal attachment portion **220E** can be fused over the whole circumference of the seal attachment portion **220E**. Also, in this battery, the area of contact between the outer surface of the seal attachment portion **220E** and the heat-sealable resin is large, when compared to the first embodiment (a case where the linear protrusions **60** are not formed in the seal attachment portion **220E**), and therefore the seal attachment portion **220E** of the valve device **200E** can be firmly fixed to the packaging material **110**.

(123) Note that the valve device **200E** is an example of the “valve device” in the present invention, and the seal attachment portion **220E** is an example of the “second portion” in the present invention. The linear protrusions **60** are each an example of a “linear protrusion” in the present invention. The air passage A3 is an example of the “air passage” in the present invention.

(124) Although the linear protrusions **60** in the sixth embodiment continuously extend over the whole circumference, the linear protrusions **60** are only required to extend in the circumferential direction, and do not necessarily have to extend over the whole circumference or be continuous. For example, in a case where the wing-shaped extended end parts **40** and **41** are provided as in the third embodiment described above, the linear protrusions **60** need not extend over the whole circumference including the wing-shaped extended end parts **40** and **41**, and a configuration is also possible in which the linear protrusions **60** are not provided in leading end parts of the wing-shaped extended end parts **40** and **41**, the linear protrusions **60** are not provided in the wing-shaped extended end parts **40** and **41**, or the linear protrusions **60** are intermittently formed in the circumferential direction.

7. Seventh Embodiment

(125) A seventh embodiment differs from the first embodiment described above in the configuration of the valve device. Other configurations are basically the same as those in the first embodiment. Here, differences from the first embodiment will be described.

(126) FIG. 20 is a plan view of a valve device 200F that is mounted in a battery according to the seventh embodiment. As shown in FIG. 20, the valve device 200F includes a valve function portion 210F and a seal attachment portion 220F. At least a portion of the seal attachment portion 220F is sandwiched between the packaging materials 110 and 120 and is subjected to heat sealing. The cross-sectional shapes of the valve function portion 210F and the seal attachment portion 220F differ from those in the first embodiment.

(127) FIG. 21 is a cross-sectional view taken along line XXI-XXI in FIG. 20. As shown in FIG. 21, a cross section of the valve function portion 210F has a semicircular shape. That is, a surface of the valve function portion 210F in the direction of the arrow U is a flat surface. Also, a cross section of the seal attachment portion 220F includes wing-shaped extended end parts 40F and 41F on both end parts in the direction indicated by the arrows L and R. A surface of the seal attachment portion 220F in the direction of the arrow U is a flat surface. The surface of the valve function portion 210F in the direction of the arrow U and the surface of the seal attachment portion 220F in the direction of the arrow U are flush with each other.

(128) Accordingly, if the valve device 200F is arranged such that the surface in the direction of the arrow U faces downward, the valve device 200F does not roll over. According to the battery of the seventh embodiment, the valve device 200F does not roll over when the valve device 200F is to be attached to the housing body 100, and therefore the valve device 200F can be easily positioned.

(129) FIG. 22 is a diagram showing a state where the valve device 200F is to be attached to the housing body 100. As shown in FIG. 22, the flat surface of the valve device 200F is placed on a surface of the innermost layer of the packaging material 120 when the valve device 200F is to be attached to the housing body 100. In this state, the valve device 200F does not roll over. Therefore, according to the battery of the seventh embodiment, the valve device 200F can be easily positioned when the valve device 200F is to be attached to the housing body 100. Also, in a state where the battery is complete, a bulge of the joined edge portion 130 that is formed by the valve device 200F can be oriented in the direction in which the housing body 100 bulges, i.e., in the upward direction in FIG. 22 in which the molded part 112 protrudes.

(130) Note that the valve device 200F is an example of the “valve device” in the present invention, the valve function portion 210F is an example of the “first portion” in the present invention, and the seal attachment portion 220F is an example of the “second portion” in the present invention. The air passage A4 is an example of the “air passage” in the present invention.

8. Variations

(131) Although the first to seventh embodiments have been described, the present invention is not limited to the first to seventh embodiments described above, and various changes can be made without departing from the gist of the present invention. The following describes variations. Note that the following variations can be appropriately combined.

(132) 8-1

(133) In the first to seventh embodiments described above, the cross-sectional shape of the seal attachment portion (e.g., the seal attachment portion 220) is based on a circular shape. However, the cross-sectional shape of the seal attachment portion is not limited to such a shape. For example, the cross-sectional shape of the seal attachment portion may also be based on a polygonal shape.

(134) FIG. 23 is a diagram showing a cross section of a valve device 200G according to a first variation. As shown in FIG. 23, a cross section of a seal attachment portion 220G of the valve device 200G has a rhombic shape. A length L7 of the seal attachment portion 220G in the width direction of the battery is longer than a length L8 of the seal attachment portion 220G in the thickness direction of the battery. In the thickness direction of the battery described above, the difference between a length of a portion of the joined edge portion 130 where the seal attachment portion 220G is sandwiched and a length of a portion of the joined edge portion 130 where the seal attachment portion 220G is not sandwiched is further reduced. Therefore, according to this battery, it is possible to appropriately apply pressure and heat to the heat-sealable resin layers 35 over the

entire periphery of the housing body **100** and appropriately fuse the mutually facing heat-sealable resin layers **35**, and accordingly, the seal attachment portion **220G** of the valve device **200G** can be firmly fixed to the housing body **100**.

(135) FIG. **24** is a diagram showing a cross section of a valve device **200H** according to a second variation. As shown in FIG. **24**, a cross section of a seal attachment portion **220H** of the valve device **200H** has a rhombic shape of which both end parts in the thickness direction of the battery are chamfered, or a hexagonal shape. A length **L9** of the seal attachment portion **220H** in the width direction of the battery is longer than a length **L10** of the seal attachment portion **220H** in the thickness direction of the battery. In the thickness direction of the battery described above, the difference between a length of a portion of the joined edge portion **130** where the seal attachment portion **220H** is sandwiched and a length of a portion of the joined edge portion **130** where the seal attachment portion **220H** is not sandwiched is further reduced. Therefore, according to this battery, it is possible to appropriately apply pressure and heat to the heat-sealable resin layers **35** over the entire periphery of the housing body **100** and appropriately fuse the mutually facing heat-sealable resin layers **35**, and accordingly, the seal attachment portion **220H** of the valve device **200H** can be firmly fixed to the housing body **100**.

(136) FIG. **25** is a diagram showing a cross section of a valve device **200I** according to a third variation. As shown in FIG. **25**, a cross section of a seal attachment portion **200I** of the valve device **200I** has a shape that is obtained by providing wing-shaped extended end parts **401** and **411** on both end parts of a rhombic shape (in the width direction of the battery). In this battery, the length in the thickness direction of the battery smoothly changes from a portion of the joined edge portion **130** where the seal attachment portion **200I** is not sandwiched to a portion of the joined edge portion **130** where the seal attachment portion **200I** is sandwiched, when compared to the first embodiment (a case where the wing-shaped extended end parts **401** and **411** are not provided in the seal attachment portion **200I**), for example. Accordingly, in this battery, an excessive force is not applied to the packaging materials **110** and **120** at the boundary between a position where the seal attachment portion **220I** is sandwiched between the heat-sealable resin layers **35** and a position where the seal attachment portion **220I** is not sandwiched between the heat-sealable resin layers **35**, and therefore the seal attachment portion **220I** of the valve device **200I** can be firmly fixed to the housing body **100**.

(137) FIG. **26** is a plan view of a valve device **200J** according to a fourth variation. As shown in FIG. **26**, the valve device **200J** includes a valve function portion **210J** and a seal attachment portion **220J**. An air passage **A5** is formed inside the seal attachment portion **220J**.

(138) FIG. **27** is a cross-sectional view taken along line XXVII-XXVII in FIG. **26**. This cross section can also be said as being a surface with respect to which a center line **C1** of the air passage **A5** is the normal line. As shown in FIG. **27**, the cross section of the seal attachment portion **220J** of the valve device **200J** has a hexagonal shape (polygonal shape). **Rs** (e.g., $R=0.2\text{ mm to }2.0\text{ mm}$) are formed at corners of the hexagonal shape. According to this battery, for example, it is possible to reduce the possibility that a portion of the seal attachment portion **220J** located in the housing body **100** will damage the battery element **400** in the housing body **100**, and reduce the possibility that a portion of the seal attachment portion **220J** sandwiched between the heat-sealable resin layers **35** will damage the heat-sealable resin layers **35** and impair the insulating performance of the heat-sealable resin layers **35**.

(139) 8-2

(140) In the first to seventh embodiments described above, the flange portion **114** of the packaging material **110** is flat. However, the shape of the flange portion **114** is not limited to such a shape. For example, a valve device arrangement portion for arranging the seal attachment portion **220** of the valve device **200** may also be formed in the flange portion **114** in advance.

(141) FIG. **28** is a plan view of a packaging material **110K** according to a fifth variation. As shown in FIG. **28**, a valve device arrangement portion **116K** is formed in the flange portion **114K**.

(142) FIG. 29 is a cross-sectional view taken along line XXIX-XXIX in FIG. 28. As shown in FIG. 29, the valve device arrangement portion **116K** formed in the flange portion **114K** has a semicircular shape. The diameter of this semicircle is slightly larger than the diameter of the seal attachment portion **220**, for example. Heat sealing is performed on the peripheral edge portion of the housing body in a state where the seal attachment portion **220** is arranged in the valve device arrangement portion **116K**, for example. As a result, deformation of the packaging material during the heat sealing process is suppressed, and it is possible to reduce the possibility that pinholes will be formed or the packaging material will break in the vicinity of the seal attachment portion **220**. Note that the valve device arrangement portion **116K** does not necessarily have to be provided in the packaging material **110K**, and may also be provided in the packaging material **120**. In this case as well, it is possible to achieve effects similar to those achieved in the case where the valve device arrangement portion **116K** is provided in the packaging material **110K**.

(143) 8-3

(144) In the first to seventh embodiments described above, only a portion of the seal attachment portion (e.g., the seal attachment portion **220**) is sandwiched between the heat-sealable resin layers **35** in the joined edge portion **130**. However, the attached state of the seal attachment portion is not limited to such a state. For example, the entire seal attachment portion may also be sandwiched between the heat-sealable resin layers **35** in the joined edge portion **130**. In this case as well, Rs are formed at corners of the end part of the seal attachment portion (e.g., the seal attachment portion **220**) on the side opposite to the valve function portion (e.g., the valve function portion **210**) in a plan view, and therefore the end part is unlikely to damage the heat-sealable resin layers **35** and impair the insulating performance of the heat-sealable resin layers **35**.

(145) 8-4

(146) In the first to seventh embodiments described above, a step is formed at the boundary between the valve function portion (e.g., the valve function portion **210**) and the seal attachment portion (e.g., the seal attachment portion **220**) of the valve device (e.g., the valve device **200**). However, the step does not necessarily have to be formed at the boundary between the valve function portion and the seal attachment portion. For example, a configuration is also possible in which the diameter of the cross section of the valve function portion is the same as the diameter of the cross section of the seal attachment portion, and the valve function portion and the seal attachment portion are continuous with each other forming a flat surface.

(147) 8-5

(148) In the first to seventh embodiments described above, the cross-sectional shape of the air passage (e.g., the air passage **A1**) formed in the seal attachment portion (e.g., the seal attachment portion **220**) is based on a circular shape. However, the cross-sectional shape of the air passage is not limited to such a shape. For example, the cross-sectional shape of the air passage may also be based on a polygonal shape.

(149) 8-6

(150) In the first to seventh embodiments described above, Rs are formed at corners of the end part of the seal attachment portion (e.g., the seal attachment portion **220**) on the side opposite to the valve function portion (e.g., the valve function portion **210**). However, Rs do not necessarily have to be formed at the corners.

(151) 8-7

(152) In the first to seventh embodiments described above, the valve device (e.g., the valve device **200**) is a return valve. However, the valve device does not necessarily have to be a return valve. The valve device may also be a breaking valve or a selective permeation valve.

(153) 8-8

(154) In the first to seventh embodiments described above, the tabs **300** are provided in both end parts of the housing body **100** in the direction indicated by the arrows L and R, and the valve device (e.g., the valve device **200**) is provided in the end part of the housing body **100** in the

direction of the arrow F (see FIG. 1 again). However, the positional relationship between the valve device **200** and the tabs **300** is not limited to such a relationship. For example, a configuration is also possible in which both of the tabs **300** are arranged on the same side of the peripheral edge portion of the housing body **100** and the valve device is arranged between the two tabs **300**, or both of the tabs **300** are arranged on the same side of the peripheral edge portion of the housing body **100** and the valve device is arranged on any of three sides other than the side on which the tabs **300** are arranged.

(155) 8-9

(156) In the first to seventh embodiments described above, the housing body **100** includes the packaging material **110** that is formed through embossing or the like and the packaging material **120** that is separate from the packaging material **110**. However, the housing body **100** does not necessarily have to have such a configuration.

(157) For example, the packaging material **110** and the packaging material **120** may also be formed as a single piece (i.e., connected) in advance so as to share a side. In this case, a configuration is also possible in which the packaging material **110** and the packaging material **120** are formed as a single piece (i.e., connected) in an end part of the flange portion **114** of the packaging material **110**, and the battery element **400** is sealed in the housing body **100** by sealing four sides in a state where the packaging materials **110** and **120** are overlaid on each other. Alternatively, a configuration is also possible in which the flange portion **114** is omitted along the side at which the packaging materials **110** and **120** are connected, and the battery element **400** is sealed in the housing body **100** by sealing three sides in a state where the packaging materials **110** and **120** are overlaid on each other.

(158) Also, the packaging material **120** may also be formed so as to have a shape similar to that of the packaging material **110**, for example. The housing body **100** may also be a pouch-type housing body, for example. The pouch-type housing body may be any of a three-side sealing type, a four-side sealing type, a pillow type, a gusset type, and the like.

(159) 8-10

(160) In the first to seventh embodiments described above, the housing of the valve function portion (e.g., the valve function portion **210**) and the housing of the seal attachment portion (e.g., the seal attachment portion **220**) are made of the same material (resin). However, the housing of the valve function portion and the housing of the seal attachment portion do not necessarily have to be made of the same material. For example, a configuration is also possible in which the housing of the valve function portion and the housing of the seal attachment portion are made of different materials, and the material of the valve function portion has a higher melting point than the material of the seal attachment portion. For example, the seal attachment portion may be made of polypropylene (PP) and the valve function portion may be made of a resin that has a higher melting point than PP (e.g., a fluorine-based resin, a polyester-based resin, a polyimide-based resin, a polycarbonate-based resin, or an acrylic resin) or metal. Fluorocarbon resins that have a good barrier property are preferably used for the seal attachment portion.

(161) In this battery, the material of the valve function portion has a higher melting point than the material of the seal attachment portion, and therefore the valve function portion is unlikely to deform due to heat even if pressure and heat are applied to the seal attachment portion when the mutually facing heat-sealable resin layers **35** are fused. Therefore, according to this battery, the valve mechanism in the valve function portion can be kept from breaking when the mutually facing heat-sealable resin layers **35** are fused.

(162) 8-11

(163) In the first to seventh embodiments described above, the housing of the valve device **200** is made of resin and the seal attachment portion **220** is directly sandwiched between the heat-sealable resin layers **35**. However, the housing of the valve device **200** does not necessarily have to be made of resin, and may also be made of metal (e.g., aluminum or stainless steel). In this case, an adhesive

protective film may also be arranged between the seal attachment portion **220** and the heat-sealable resin layers **35**. The adhesive protective film is configured such that one surface adheres at least to resin and the other surface adheres at least to metal. Various known adhesive protective films can be employed as the adhesive protective film, and an adhesive protective film that is the same as the tab films **310** can be used, for example.

(164) 8-13

(165) In the first to seventh embodiments described above, Rs are formed on the outer peripheral side of the seal attachment portion (e.g., the seal attachment portion **220**, i.e., at corners of the end part of the seal attachment portion on the side opposite to the valve function portion (e.g., the valve function portion **210**)), but Rs are not formed on the inner peripheral side of the seal attachment portion (i.e., in an end part of the air passage (e.g., the air passage **A1**)). However, Rs may also be formed on the inner peripheral side of the seal attachment portion. If Rs are formed on the inner peripheral side of the seal attachment portion, it is possible to reduce the possibility that corners on the inner peripheral side of the seal attachment portion will be chipped and chips (e.g., pieces of resin, metal, etc.) will fall into the housing body **100**.

(166) 8-13

(167) In the seventh embodiment described above, flat surfaces are formed in outer surfaces of both the valve function portion **210F** and the seal attachment portion **220F** (see FIG. **21** again). However, it is not always necessary to form flat surfaces in the outer surfaces of both the valve function portion **210F** and the seal attachment portion **220F**. It is sufficient to form a flat surface in the outer surface of at least one of the valve function portion **210F** and the seal attachment portion **220F**.

(168) 8-14

(169) The battery **10** according to the first to seventh embodiments described above is a secondary battery, but is defined as being what outputs electricity, and accordingly encompasses power storage devices such as capacitors, electric double layer capacitors (EDLCs), and lithium ion capacitors. Also, the type of secondary battery is not specifically limited, and examples of the secondary battery include lithium ion batteries, lithium ion polymer batteries, lead storage batteries, nickel-hydrogen storage batteries, nickel-cadmium storage batteries, nickel-iron storage batteries, nickel-zinc storage batteries, silver oxide-zinc storage batteries, metal-air batteries, multivalent cation batteries, and solid-state batteries.

(170) 8-15

(171) In the first to seventh embodiments described above, an adhesive film that is configured to adhere to both the seal attachment portion (e.g., the seal attachment portion **220**) and the packaging materials **110** and **120** may also be arranged between the seal attachment portions **220** and **220A** to **220J** and the packaging materials **110** and **120**. Such an example will be described below in detail.

(172) FIG. **30** is a plan view of a valve device **200K**. FIG. **31** is a cross-sectional view taken along line XXXI-XXXI in FIG. **30**. As shown in FIGS. **30** and **31**, the valve device **200K** includes a valve function portion **210K**, a seal attachment portion **220K**, and an adhesive film **600**. That is, the adhesive film **600** that corresponds to an adhesive member is attached to the seal attachment portion **220K** of the valve device **200K** in advance. The valve function portion **210K** and the seal attachment portion **220K** are made of metal. The valve function portion **210K** and the seal attachment portion **220K** are made of aluminum, brass, stainless steel, or the like. Note that the valve function portion **210K** and the seal attachment portion **220K** do not necessarily have to be made of metal, and may also be made of resin, for example.

(173) The adhesive film **600** is configured to adhere to both the seal attachment portion **220K** and the heat-sealable resin layers **35** (FIG. **4**) of the packaging materials **110** and **120** through heat sealing. Various known adhesive films can be employed as the adhesive film **600**. For example, the adhesive film **600** may be a single-layer film of maleic anhydride-modified polypropylene (PPa) or a laminate film of PPa, polyethylene naphthalate (PEN), and PPa. Also, resins that can adhere to

metal, such as an ionomer resin, modified polyethylene, and EVA, can also be employed instead of the PPA resin described above.

(174) In the present embodiment, a laminate film that has a three-layer structure including a core material and is constituted by PPA/PEN (core material)/PPA is employed as the adhesive film **600**. Other than PEN, various known materials can also be employed as the core material. For example, the core material may also be polyester fibers, polyamide fibers, or carbon fibers.

(175) The adhesive film **600** adheres to the seal attachment portion **220K** in a state of covering the outer peripheral surface of the seal attachment portion **220K**. As described above, the adhesive film **600** also adheres to the heat-sealable resin layers **35** of the packaging materials **110** and **120**, and accordingly, even if the seal attachment portion **220K** is made of metal, the valve device **200K** and the packaging materials **110** and **120** can be easily bonded through heat sealing. Also, even if the seal attachment portion **220K** is made of resin, the adhesive film **600** securely adheres to the heat-sealable resin layers **35** of the packaging materials **110** and **120**, and accordingly, the valve device **200K** and the packaging materials **110** and **120** can be securely bonded through heat sealing. The adhesive film **600** more effectively functions particularly in a case where the seal attachment portion **220K** is made of Teflon (registered trademark) among resins. Note that a cross section taken along line XXXI-XXXI in FIG. **30** has a teary eye shape (a shape that includes a circular portion **613** and wing-shaped extended end parts **614** and **615**). That is, the entire periphery of the shape of the cross section taken along line XXXI-XXXI in FIG. **30** is curved. More specifically, the outer periphery of the circular portion **613** is outwardly curved in the cross section, and both end sides of the circular portion **613** are smoothly connected to base end sides of the wing-shaped extended end parts **614** and **615** with the outer periphery inwardly curved in the cross section, and therefore the adhesive film **600** can be bonded to the outer periphery of the seal attachment portion **220K** with no gap formed therebetween.

(176) Also, a length **W1** of the adhesive film **600** in the width direction is longer than a length **W2** of the seal attachment portion **220K** in the width direction. That is, in the valve device **200K**, the adhesive film **600** extends past the outer periphery of the seal attachment portion **220K** (FIG. **31**). Therefore, according to the valve device **200K**, the valve device **200K** and the packaging materials **110** and **120** can be more securely bonded since the adhesive film **600** is arranged over a wide range.

(177) Also, the arrangement range of the adhesive film **600** reaches the lower end of the seal attachment portion **220K**. The reason for this will be described. As described above, a secondary battery is housed in the housing body **100** to which the valve device **200K** is attached. In this case, if the seal attachment portion **220K** (metal) is exposed in a wide range, an electrode of the secondary battery and the seal attachment portion **220K** are likely to come into contact with each other, and a short circuit is likely to occur. In the valve device **200K**, the arrangement range of the adhesive film **600** reaches the lower end of the seal attachment portion **220K**. Therefore, according to the valve device **200K**, it is possible to reduce the possibility that the valve device **200K** will cause a short circuit. Note that the adhesive film **600** may also be arranged so as to extend downward past the lower end of the seal attachment portion **220K**.

(178) 8-16

(179) In the first to seventh embodiments described above, helium leakage amounts of the valve devices (the valve devices **200** and **200A** to **200J**) are not particularly described. The helium leakage amount of each valve device may be as described below, for example. The following describes the helium leakage amount of the valve device **200** as a representative example of the valve devices. Note that the helium leakage amount of the valve device **200** can also be applied as helium leakage amounts in the other embodiments (the second to seventh embodiments).

(180) As described above, the valve device **200** is configured to discharge gas from the inside to the outside of the housing body **100** if the internal pressure of the housing body **100** has reached or exceeded a predetermined value due to gas generated in the housing body **100**. If sealing

performance of the valve device **200** is excessively high, the valve device **200** may fail to function even if the internal pressure of the housing body **100** has reached or exceeded the predetermined value. On the other hand, if sealing performance of the valve device **200** is lower than required, it is highly likely that water vapor (moisture) will enter the housing body **100** from the external environment during a normal period (when the internal pressure of the housing body **100** is lower than the predetermined value).

(181) In the valve device **200** according to the present embodiment, the helium leakage amount of the valve device **200** is adjusted to simultaneously realize high sealing performance of the valve device **200** and effective suppression of the intrusion of water vapor into the housing body **100**.

(182) The inventor(s) of the present invention found that if the helium leakage amount from a secondary side to a primary side of the valve device **200** measured in an environment at 25° C. in accordance with a method defined in “vacuum spraying method” of JIS Z2331:2006 “Method for helium leak testing” is from 5.0×10^{-11} Pa·Math.m.sup.3/sec to 5×10^{-6} Pa·Math.m.sup.3/sec inclusive, it is possible to simultaneously realize high sealing performance of the valve device **200** and effective suppression of the intrusion of water vapor into the housing body **100**. Therefore, the helium leakage amount of the valve device **200** may be from 5.0×10^{-11} Pa·Math.m.sup.3/sec to 5×10^{-6} Pa·Math.m.sup.3/sec inclusive when measured in an environment at 25° C. using the method defined in the above-described standard. Note that the secondary side of the valve device **200** refers to the outside of the housing body **100** when the valve device **200** is attached to the housing body **100**. Also, the primary side of the valve device **200** refers to the inside of the housing body **100** when the valve device **200** is attached to the housing body **100**.

(183) The upper limit of the helium leakage amount of the valve device **200** is preferably about 4.5×10^{-6} Pa·Math.m.sup.3/sec or less, more preferably about 1.0×10^{-6} Pa·Math.m.sup.3/sec or less, further preferably about 1.0×10^{-7} Pa·Math.m.sup.3/sec or less, and yet more preferably about 1.0×10^{-8} Pa·Math.m.sup.3/sec or less, for example, and the lower limit of the helium leakage amount of the valve device **200** is 5.0×10^{-11} Pa·Math.m.sup.3/sec or more. Preferable ranges of the helium leakage amount of the valve device **200** are from 5.0×10^{-11} Pa·Math.m.sup.3/sec to about 4.5×10^{-6} Pa·Math.m.sup.3/sec, from 5.0×10^{-11} Pa·Math.m.sup.3/sec to about 1.0×10^{-6} Pa·Math.m.sup.3/sec, from 5.0×10^{-11} Pa·Math.m.sup.3/sec to about 1.0×10^{-7} Pa·Math.m.sup.3/sec, and from 5.0×10^{-11} Pa·Math.m.sup.3/sec to about 1.0×10^{-8} Pa·Math.m.sup.3/sec, for example.

(184) If the helium leakage amount satisfies the upper limit described above, the intrusion of water vapor (moisture) from the external environment into the housing body **100** can be effectively suppressed. Also, if the helium leakage amount satisfies the lower limit described above, gas generated in the housing body **100** can be discharged to the outside. Note that if the helium leakage amount is too small, it is difficult to stably discharge gas generated in the housing body **100** to the outside of the housing body **100**. Also, if a battery cell is continuously used with such a valve device being not opened for a long period, it is highly likely that the valve device will not appropriately open even if the internal pressure has increased to a design value.

(185) Furthermore, if the helium leakage amount of the valve device **200** is set to a range from 5.0×10^{-11} Pa·Math.m.sup.3/sec to about 2.0×10^{-10} Pa·Math.m.sup.3/sec or from 5.0×10^{-11} Pa·Math.m.sup.3/sec to about 1.5×10^{-10} Pa·Math.m.sup.3/sec, the intrusion of water vapor (moisture) from the external environment into the housing body **100** can be particularly effectively suppressed. In order to set the helium leakage amount as described above, the shape of a portion in which a valve seat of the valve mechanism and the ball come into contact with each other needs to be designed and processed with very high precision that is not required in conventional check valves, as described later.

(186) Note that the helium leak testing is carried out as follows. That is, in the helium leak testing, the helium leakage amount from the secondary side to the primary side of the valve device **200** is

measured in accordance with the method defined in “vacuum spraying method” of JIS Z2331:2006 “Method for helium leak testing”. Specifically, a helium leak detector is used as a test apparatus. Also, a gas valve (the valve function portion **210**) of the valve device **200** is set in a leak test jig (a jig for which it was confirmed that helium does not leak when a dummy valve device including a closed gas valve was set), and the jig is set in the helium leak detector via a test port. It is confirmed that helium does not leak between the jig and the helium leak detector. Thereafter, the valve device **200** is evacuated to 13 Pa from the primary side, a 99.99% helium gas is sprayed from the secondary side of the valve device **200**, and measurement is started. Evaluation results are recorded with a spraying period set to 1 to 2 seconds and a waiting period set to 2 to 4 seconds. Note that it is also possible to cover the same valve device **200** with a hood having a capacity of 50 ml and wait for 20 seconds in accordance with the method defined in “vacuum covering method (vacuum hood method)” of JIS Z2331:2006 “Method for helium leak testing” to make it sure that similar measurement results can be obtained. In any case, measurement is performed in an environment at 25° C. The lower limit of a pressure difference between the (187) primary side and the secondary side of the valve device **200** (i.e., an opening pressure of the valve device **200**) is preferably about 0.05 MPa or more, and more preferably about 0.1 MPa or more, for example. The upper limit of the pressure difference is preferably about 1 MPa or less, and more preferably about 0.3 MPa or less, for example. Preferable ranges of the pressure difference are about 0.05 to 1 MPa, about 0.05 to 0.3 MPa, about 0.1 to 1 MPa, and about 0.1 to 0.3 MPa, for example. If the pressure difference is as described above, gas generated in the housing body **100** can be appropriately discharged to the outside, and the intrusion of water vapor (moisture) from the external environment can be effectively suppressed.

(188) The internal pressure of the battery **10** (the housing body **100**) to which the valve device **200** has been attached is preferably set to be no higher than a predetermined pressure. Although the setting value of the internal pressure is appropriately determined according to the type of a package including the valve device, the setting value is preferably about 0.1 MPa or less, and more preferably about 1.0×10^{-2} MPa or less. The lower limit of the setting value is about 1.0×10^{-10} MPa or more, for example. Preferable ranges of the internal pressure are about 1.0×10^{-10} to 0.1 MPa and about 1.0×10^{-10} to 1.0×10^{-2} MPa, for example.

(189) The helium leakage amount of the valve device **200** can be set using a known method. For example, the helium leakage amount can be adjusted by designing materials, shapes, and sizes of members (e.g., the ball **214**, the o ring **212**, the spring **216**, and the exhaust port **O1**) constituting the valve function portion **210** of the valve device **200** and a force with which the ball **214** is pressed by the spring **216**.

(190) For example, if an elastic body is used as one of the ball **214** and the O ring **212** of the valve mechanism and a member that has a high degree of hardness, such as metal, is used as the other, the helium leakage amount can be easily set to the range from 5.0×10^{-11} Pa.Math.m.sup.3/sec to 5.0×10^{-6} Pa.Math.m.sup.3/sec inclusive. In order to reduce the helium leakage amount, it is effective to use elastic bodies for both of the ball **214** and the O ring **212** of the valve mechanism, but if the helium leakage amount is too small, it is difficult to appropriately discharge gas generated in the housing body **100** to the outside as described above, and therefore, materials, shapes, sizes, and the like of the members constituting the valve mechanism are appropriately adjusted. For example, if a portion of the O ring **212** that comes into contact with the ball **214** in the valve mechanism has a shape that conforms to the shape of the surface of the ball **214**, the helium leakage amount can be easily designed to fall within the above-described ranges.

(191) That is, in order to set the helium leakage amount of the valve device **200** to the range from 5.0×10^{-11} Pa.Math.m.sup.3/sec to about 2.0×10^{-10} Pa.Math.m.sup.3/sec or from 5.0×10^{-11} Pa.Math.m.sup.3/sec to about 1.5×10^{-10} Pa.Math.m.sup.3/sec, the shape of the portion in which the O ring **212** and the ball **214** of the valve mechanism come into contact with each other needs to be designed and processed with very high precision that is not required in

conventional check valves. For example, it is effective to make the ball **214** and a portion of the O ring **212** that comes into contact with the ball **214** have an average surface roughness of 20 μm or less, preferably 5 μm or less, and more preferably 1 μm or less. However, there is a problem in that the valve device **200** may fail to appropriately operate (the valve function portion **210** may fail to open) if members that are too precise are brought into contact with each other, and therefore the surface roughness needs to be adjusted such that the helium leakage amount falls within the above-described ranges.

(192) 8-17

(193) In the first to seventh embodiments described above, a maximum distortion in the thickness direction of the housing body **100** after gas is discharged from the inside to the outside of the housing body **100** via the valve device (which hereinafter will also be simply referred to as the “maximum distortion”) is not particularly described. The maximum distortion in the thickness direction of the housing body **100** may be as described below, for example. The following describes, as a representative example, the maximum distortion of the housing body **100** in the first embodiment described above. Note that the maximum distortion of the housing body **100** in the first embodiment can also be applied as the maximum distortion of the housing body **100** in the other embodiments (the second to seventh embodiments).

(194) The battery **10** includes the housing body **100** and the valve device **200**. The housing body **100** is constituted by at least one laminate including at least the base material layer, the barrier layer, and the heat-sealable resin layer that are layered in that order. The housing body **100** houses the battery element **400**. The valve device **200** is in communication with the inside of the housing body **100**. The valve device **200** is configured to reduce the internal pressure of the housing body **100** if the internal pressure is increased due to gas generated in the housing body **100**. The maximum distortion of the thickness of the housing body **100** (the packaging materials **110** and **120**) after gas is discharged from the inside to the outside of the housing body **100** via the valve device **200** may be less than 30%. That is, the opening pressure of the valve device **200** of the battery **10** may be set such that the maximum distortion of the thickness is less than 30%.

(195) As a result of the battery **10** having such a feature, gas is discharged via the valve device **200** at appropriate timings, and therefore it is possible to suppress the occurrence of problems such as the formation of large creases in the housing body **100** after gas is discharged and large deformation of the housing body **100** after gas is discharged.

(196) As described above, focusing on the relationship between the maximum distortion of the housing body **100** and the opening pressure of the valve device **200**, the maximum distortion of the housing body **100** of the battery **10** can be set to be less than 30%. As a result, in the battery **10**, gas can be discharged via the valve device **200** to the outside of the housing body **100** at appropriate timings before problems occur such as the formation of large creases in the housing body **100** after gas is discharged and large deformation of the housing body **100** after gas is discharged. That is, when the internal pressure of the housing body **100** is increased, if a change in the thickness of the housing body **100** is observed and the opening pressure of the valve device **200** is set such that the valve device **200** will open before the thickness is reduced by 30%, it is possible to effectively suppress problems such as the formation of large creases in the housing body **100** after the valve device **200** is open and large deformation of the housing body **100** after the valve device **200** is open.

(197) The upper limit of the maximum distortion is preferably about 28% or less, and more preferably 27% or less, for example. Also, the lower limit of the maximum distortion is preferably about 2% or more, and more preferably about 4% or more, for example. If the maximum distortion is set as described above, gas can be discharged to the outside of the housing body **100** at more appropriate timings before problems occur such as the formation of large creases in the housing body **100** after gas is discharged and large deformation of the housing body **100** after gas is discharged. Preferable ranges of the maximum distortion are about 2% or more and less than 30%,

and from about 4% to about 28%, for example.

(198) The maximum distortion of the housing body **100** is measured as described below. First, the same two housing bodies **100** are prepared as targets of measurement of the maximum distortion. Next, lines are drawn in a lattice pattern at intervals of 1 mm on outer surfaces of the housing bodies **100**. At this time, the lines are drawn at the same positions on the respective outer surfaces of the two housing bodies **100**. Next, with respect to one of the housing bodies **100**, the valve device **200** is sealed and another air passage is provided in the housing body **100**, or the valve function is removed from the valve device **200** so that the valve device **200** functions as the air passage, and air is introduced via the air passage to the inside of the housing body **100** to increase the internal pressure up to 1 MPa and cause the housing body **100** to swell. Next, air is removed via the air passage to reduce the internal pressure to normal pressure, the housing body **100** is cut along a line of the above-described lattice pattern, and the thickness of the cross section is measured. The housing body **100** into which air has not been introduced is also cut along a line of the above-described lattice pattern, and the thickness of the cross section is measured. Next, a position at which the thickness decreased the most when compared to the thickness at the same position of the housing body **100** to which air has not been introduced is taken to be a position with the maximum distortion, and the ratio (%) of reduction in the thickness at the position with the maximum distortion is taken to be the maximum distortion (%). For example, if the housing body **100** of which the internal pressure has not been increased has a thickness of 100 μm at the position with the maximum distortion and the housing body **100** of which the internal pressure has been increased has a thickness of 70 μm at the same position, the maximum distortion is 30%.

LIST OF REFERENCE NUMERALS

(199) **10** Battery **31** Base material layer **32** Adhesive agent layer **33** Barrier layer **34** Adhesive layer **35** Heat-sealable resin layer **40, 401, 41, 411** Wing-shaped extended end part **50, 51** Pillar **60** Linear protrusion **100** Housing body **110, 110K, 120** Packaging material **112** Molded part **114, 114K** Flange portion **116K** Valve device arrangement portion **130** Joined edge portion **200, 200A, 200B, 200C, 200D, 200E, 200F, 200G, 200H, 200I, 200J, 200K** Valve device **210, 210A, 210B, 210C, 210E, 210F, 210G, 210H, 210I, 210J, 210K** Valve function portion **212** O ring **214** Ball **216** Spring **218** Membrane **220, 220A, 220B, 220C, 220D, 220E, 220F, 220G, 220H, 220I, 220J, 220K** Seal attachment portion **300** Tab **310** Tab film **400** Battery element **600** Adhesive film **613** Circular portion **614, 615** Wing-shaped extended end part **A1, A2, A3, A4, A5, A6** Air passage **C1** Center line **O1** Exhaust port

Claims

1. A battery comprising: a battery element; a housing body that is constituted by at least one laminate including at least a base material layer, a barrier layer, and a heat-sealable resin layer layered in that order and houses the battery element; and a valve device that is in communication with the inside of the housing body, wherein sections of the heat-sealable resin layer of the at least one laminate face each other in a peripheral edge portion of the housing body, a joined edge portion in which the mutually facing sections of the heat-sealable resin layer of the at least one laminate are fused together is formed in the peripheral edge portion of the housing body, the valve device includes: a first portion in which a valve mechanism is formed, the valve mechanism being configured to reduce an internal pressure of the housing body if the internal pressure is increased due to gas generated in the housing body; and a second portion in which an air passage is formed, the air passage being configured to guide gas generated in the housing body toward the valve mechanism, the first portion is located on an outer side of an outer edge of the joined edge portion, at least a portion of the second portion is sandwiched between the sections of the heat-sealable resin layer of the at least one laminate in the joined edge portion, a length of the second portion in a width direction of the battery is longer than a length of the second portion in a thickness direction

of the battery, and wherein a length of the air passage of the second portion in a width direction of the battery is longer than a length of the air passage of the second portion in a thickness direction of the battery, where the width direction is transverse to a longitudinal axis of the second portion.

2. A battery comprising: a battery element; a housing body that is constituted by at least one laminate including at least a base material layer, a barrier layer, and a heat-sealable resin layer layered in that order and houses the battery element; and a valve device that is in communication with the inside of the housing body, wherein sections of the heat-sealable resin layer of the at least one laminate face each other in a peripheral edge portion of the housing body, a joined edge portion in which the mutually facing sections of the heat-sealable resin layer of the at least one laminate are fused together is formed in the peripheral edge portion of the housing body, the valve device includes: a first portion in which a valve mechanism is formed, the valve mechanism being configured to reduce an internal pressure of the housing body if the internal pressure is increased due to gas generated in the housing body; and a second portion in which an air passage is formed, the air passage being configured to guide gas generated in the housing body toward the valve mechanism, the first portion is located on an outer side of an outer edge of the joined edge portion, at least a portion of the second portion is sandwiched between the sections of the heat-sealable resin layer of the at least one laminate in the joined edge portion, a length of the second portion in a width direction of the battery is longer than a length of the second portion in a thickness direction of the battery, an outer contour of the joined edge portion by the heat-sealable resin layers at a peripheral joint in contact with a portion of the second portion includes a straight line, and wherein a length of the air passage of the second portion in a width direction of the battery is longer than a length of the air passage of the second portion in a thickness direction of the battery, where the width direction is transverse to a longitudinal axis of the second portion.
